Shiva Sai Reddy Bommireddy

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EDUCATION

Indian Institute of Technology(IIT) Bhilai **2019-2023**
B.Tech, Computer Science & Engineering

EXPERIENCE

Singularium Technologies | **Software Engineer | FDE, Platform & Infra** | Bengaluru | Apr 2025 - present

- **Owned Django + Flask APIs + React UI on Gunicorn + Nginx service** for OCR, GPT extraction, and doctor interpretation across **23 insurers at 10K+ PHI docs/day**; Terraform on AWS, GitHub Actions blue-green, per-insurer isolation, KMS, token-bucket rate limits on Azure DI + GPT, OTel via ADOT; **p95 <200ms API/<30s end-to-end, 99.9% uptime.**
- Dockerized Lambda + Step Functions QC pipeline validating OCR/GPT fields against **200+ versioned insurer rules**, **auto-clears 81% of 10K+ docs/day**, routes rest to HITL; canary aliases, DLQs, CloudWatch SLOs.
- Self-hosted **DOTS OCR on vLLM/L4 GPUs**, tuning batching, PagedAttention, and GPU memory for **12 pages/sec at <800ms p95;** migrated from Azure DI, cutting OCR spend $\sim 65\%$ **across 40+ doc types.**
- Content-addressed cache (**SHA-256 + Redis**) fronting Azure DI + GPT, keyed by (doc_hash, pipeline_version) for versioned invalidation; **34% hit rate on 10K+ docs/day, p95 <40ms vs.** $\sim 22s, \sim \$9K/mo$ **saved.**
- Dual-DB: **RDS MySQL (Multi-AZ, RDS Proxy,** $\sim 8M$ **rows)** for claims/reviewers/audits; DynamoDB keyed insurer_id#doc_hash with GSIs on claim_id + uploaded_at; **sub-5ms p99 at 400 req/s across 25M+ items.**

Novaluna Technologies | **Software Developer Intern - Founding Team** | Dec 2023 - Aug 2024

- Built **Node.js + Next.js** patient coordination platform for **120+ providers across 15+ clinics**, **TypeORM transactions** on PostgreSQL, **SQS**-queued calendar webhooks, **Socket.io** for real-time appointment updates to a React provider portal.
- Reduced appointment latency from **420ms to <25ms across 1.2M+ rows** using **composite B-tree indexes** on (provider_id, appointment_date, status); exposed **JWT-scoped REST APIs** for provider/patient roles.
- Built an **OpenAI function calling** triage chatbot with **Zod-validated structured outputs** and confidence-gated human handoff, auto-routed **68%** of **2K+ daily patient queries across 14 clinical intents.**
- Prevented double-booking during concurrent calendar webhooks using Redis distributed locks (**SET NX + TTL**) around TypeORM transactions; **zero race-condition incidents** across **15K+ booking attempts.**

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL

Backend & Frameworks: Django, Flask, Node.js, Express.js, React.js, Next.js, Vite, Tailwind CSS, TanStack Query, SQLAlchemy, TypeORM, REST APIs, GraphQL, , WebSockets(Socket.io)

Cloud & Infra: AWS, Terraform, Docker, Nginx, Gunicorn, GitHub Actions, OpenTelemetry (ADOT), Prometheus, Grafana

Databases/Caching/Search: PostgreSQL, MySQL, DynamoDB, MongoDB, Redis, Elasticsearch, OpenSearch, Kafka

ML/AI & Document Processing: vLLM, Ollama, Qwen2.5-VL, DOTS OCR, Azure Document Intelligence, OpenAI function-calling

Security/Testing: JWT, KMS, PyTest, Jest, Playwright, k6

PROJECTS

Shopping Cart System | FastAPI, React, SQLAlchemy, Razorpay, OpenSearch, PostgreSQL, Redis, Kafka, Kubernetes | **Github**

- Built a **FastAPI + React (TypeScript)** e-commerce platform with **PostgreSQL** (products, cart, orders, **JWT auth, Razorpay payments, Shiprocket shipping):** row-level locking that prevents overselling under concurrent checkouts, **Kafka event** processing via a transactional outbox, **OpenSearch** faceted search and **Redis caching** that cut catalog **p95 latency from 99 ms to 5.5 ms at 1,000 req/s;**containerised with Docker, with a Helm chart for Kubernetes.

Docflow AI | Python, Django, PostgreSQL, Redis, React, AWS, Terraform, Prometheus/Grafana, Ollama/vLLM (Qwen) | **Github**

- Built a multi-tenant receipt-processing platform **(Django, PostgreSQL, Redis, React, self-hosted Qwen models)** that auto-approves a receipt only when its fields check out against the receipt text and its own arithmetic. On **200** labelled receipts it auto-approved **52%** with zero wrong totals, and eevaluation-driven fixes raised total-amount accuracy from **49% to 96%** on the second dataset; includes an **AWS pipeline** in **Terraform (Step Functions, Lambda)** and **Grafana monitoring.**

ACHIEVEMENTS AND CERTIFICATIONS

- Secured **AIR 4597 rank** in **JEE Advanced 2019** among 1.2 million candidates.
- Honored with the **GEET AMAL GHOSH ROY AWARD** IIT Bhilai, 2019.
- 1000+ challenges solved in various platforms like **Leetcode, Hackerrank and other coding platforms.**